

RESEARCH PAPER

Nationwide survey of patients in Japan with Bickerstaff brainstem encephalitis: epidemiological and clinical characteristics

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► An additional figure is published online only. To view this file please visit the journal online (<http://dx.doi.org/10.1136/jnnp-2012-303060>).

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Received 25 April 2012

Revised 2 July 2012

Accepted 3 July 2012

Published Online First
31 July 2012

ABSTRACT

Background Information regarding the epidemiological background of Bickerstaff brainstem encephalitis (BBE) is limited.

Methods We conducted a nationwide survey of BBE in the Japanese population in two steps: the first aimed to identify patients with brainstem encephalitis for the specified 3 year period and the second to evaluate whether the clinical picture met our diagnostic criteria for BBE.

Results The number of patients with brainstem encephalitis was estimated as 704 (95% CI 478 to 930) over the 3 years. The annual onset of BBE was roughly estimated as 100 cases, which accounted for 43% of brainstem encephalitis. BBE was slightly male predominant and often young onset. Among brainstem encephalitis patients, BBE was characterised by antecedent infectious symptoms, oropharyngeal palsy and sensory disturbance at the distal extremities with absent or decreased tendon reflexes, in addition to a triad of symptoms (external ophthalmoplegia, ataxia and impaired level of consciousness) and shorter duration to the peak, with good outcome. Anti-GQ1b antibodies were present in 75% of cases. Several BBE patients with atypical clinical features or without anti-GQ1b antibodies were also identified. These cases often had marked CSF pleocytosis, abnormal brain MRI findings and a longer duration to peak symptoms, sometimes with considerable residual deficits.

Conclusions BBE is a rare disorder but accounts for a major proportion of brainstem encephalitis. BBE consists of typical and atypical cases. Typical BBE has similar neurological and serological features to Fisher syndrome and shows good recovery whereas atypical BBE is characterised by delayed recovery, negative anti-GQ1b antibodies, and abnormal CSF and brain MRI findings with other possible pathogeneses.

INTRODUCTION

Bickerstaff brainstem encephalitis (BBE) is characterised by acutely progressive bilateral ophthalmoparesis and ataxia with impaired consciousness or pyramidal signs, or both, all of which are followed by a monophasic course with good recovery.^{1–3} BBE derives from eight original patients reported by Bickerstaff and Cloake in 1951⁴ and Bickerstaff in 1957,⁵ who speculated that their patients had an aetiology of direct viral infection of the CNS or an inflammatory reaction triggered by a systemic infectious agent. Based on neurological findings

presumably resulting from disturbed brainstem function and pathological changes in the brainstem,^{3–5} BBE is considered a phenotype of brainstem encephalitis with specific neurological deficits. However, mostly because of its rarity, it is difficult to clarify the epidemiological nosological positions of BBE among brainstem encephalitis with various aetiologies such as direct infection, demyelination due to autoimmunity or post-infectious reaction, and vasculitis. Therefore, no study has satisfactorily provided information regarding what is different in BBE from other types of brainstem encephalitis and, perhaps more importantly, its annual incidence and incidence among other forms of brainstem encephalitis. Moreover, the therapeutic efficacy of any treatment of BBE has not yet been established.⁶

Our understanding was markedly increased by the detection of autoantibodies against GQ1b ganglioside in the serum of BBE patients.⁷ The existence of anti-GQ1b antibodies is a feature in common with patients having Fisher syndrome (FS), suggesting that BBE forms a continuous spectrum with FS. Clinical, laboratory and electrophysiological analyses of BBE patients support this hypothesis.^{8,9} However, it should be noted that a considerable number of BBE patients who are negative for anti-GQ1b antibodies have also been reported: the frequency of BBE patients positive for these anti-GQ1b antibodies is reported as 65% and 68%,^{3,9} being lower than the 83–100% frequency in those having FS.^{9–11} This suggests that BBE is pathologically more heterogeneous than FS, and that some proportion of BBE patients do not fit the ‘anti-GQ1b antibody syndrome’,^{12,13} leading to the variety of clinical presentations in BBE.

Relatively large case series of BBE patients, who had been referred to a single Japanese institute for antiganglioside antibody testing, have been reported,^{3,9} but it was not possible in such a study setting to determine the epidemiological features and nosological position of BBE among brainstem encephalitis. We therefore planned a nationwide survey of BBE in the Japanese population.

METHODS

Outline of the survey

The survey was conducted according to the *Manual for nationwide epidemiological surveys of patients with intractable diseases* (2nd edition 2006, Ministry of Health, Labour, and Welfare of Japan),¹⁴ as

described below. The survey was conducted in two steps: the first step aimed to identify patients with brainstem encephalitis, irrespective of whether or not the diagnosis was BBE. The second aimed to collect clinical and laboratory information regarding the identified patients. This method allowed us to independently evaluate whether the clinical picture met the original diagnostic criteria for BBE in each individual case. This method was also helpful in detecting the clinical characteristics of patients with BBE among those with brainstem encephalitis. The study protocol was approved by the institutional ethics committee at Yamaguchi University Hospital.

Questionnaires

In the first step of the survey, a questionnaire was issued to determine the number of patients with brainstem encephalitis who had visited the institutes in the 3 years between October 2006 and September 2009. Diagnostic criteria for brainstem encephalitis, which were developed for this study, were: (1) encephalitis or encephalopathy in which a brainstem lesion was suspected neurologically or radiologically and (2) lacking symptoms or findings suggestive of cerebral hemisphere lesions. Specific symptoms or findings for (1) were exemplified in the questionnaire by conscious disturbance, double vision or external ophthalmoplegia, pyramidal signs, hemisensory disturbance, alternate hemiplegia and acute brainstem lesions on CT or MRI, and those for (2) were absence of higher brain dysfunction (eg, aphasia and apraxia) or acute lesions of the cerebral hemispheres on CT or MRI. The hospitals included in the study were randomly selected from a directory of all registered hospitals throughout Japan. Selection was made according to a stratification based on the number of beds in each hospital whereby a higher number of beds resulted in a higher probability of being selected, as performed elsewhere.^{15 16} The sampling rates were approximately 5%, 10%, 20%, 40%, 80% and 100% for general hospitals with 20–99 beds, 100–199 beds, 200–299 beds, 300–399 beds, 400–499 beds and 500 or more beds, respectively. All university hospitals, as well as neurology training hospitals where BBE patients were considered to be treated, were also surveyed. The questionnaire was sent to 3524 departments: neurology (n=1004), internal medicine (n=986), paediatrics (n=799) and neurosurgery (n=735). Estimation of patients with brainstem encephalitis was done based on the assumption that the responses of the departments were independent of the frequency of the patients. The formulae, as described elsewhere,^{15 16} were used.

In the second step of the survey, we requested details of individual patients from each department that had one or more patients with brainstem encephalitis. The questionnaire was issued to determine sex, age, diagnosis at the department, clinical features (antecedent infectious symptoms, neurological findings at nadir and Hughes disability scale at nadir and last visit), antiganglioside antibodies, results of CSF, electroencephalography, nerve conduction study, brain MRI analyses and therapies. Date of birth was also asked to exclude possible duplicate cases.

Diagnostic criteria for BBE

Diagnostic criteria for BBE were developed after discussion with five neurologists (MK, SK, KK, TKo and TKa) (box 1). 'Definite BBE' was characterised by the presence of IgG anti-GQ1b antibodies and a typical clinical picture of acute progression of the neurological triad (external ophthalmoplegia, ataxia and impaired level of consciousness). In contrast, atypical or incomplete neurological findings were admitted for diagnosis of

Box 1 Diagnostic criteria for Bickerstaff brainstem encephalitis

'Definite' Bickerstaff brainstem encephalitis is defined when (1), (2) and (4) are satisfied.

'Probable' Bickerstaff brainstem encephalitis is defined when (1) and (4), or when (2), (3) and (4) are satisfied.

- (1) Acute progressive external ophthalmoplegia,* ataxia and impaired conscious level by 4 weeks, followed by spontaneous recovery within 12 weeks after onset.
- (2) Positive for serum IgG anti-GQ1b antibodies.
- (3) Incomplete agreement on (1) because of one or more of the following reasons†
 - It is impossible to evaluate ataxia because of severe limb weakness or consciousness disturbance.
 - Unconfirmed recovery of the symptoms.
 - Remarkable laterality of external ophthalmoplegia.
 - Long tract sign (hemisensory disturbance, pyramidal sign or spasticity) instead of impaired level of consciousness.
- (4) Other conditions are excluded in laboratory and image tests: The excluded conditions are Wernicke encephalopathy, cerebrovascular disorder, multiple sclerosis, neuromyelitis optica, neuro-Behçet syndrome, neuro-Sweet disease, pituitary apoplexy, viral brainstem encephalitis, myasthenia gravis, brainstem tumour, vasculitis, botulism, Hashimoto encephalopathy.

*Lateral symmetry is the rule but mild laterality is also permitted.

†Features other than the incomplete item(s) must meet (1).

'probable BBE', if anti-GQ1b antibodies were positive. Similarly, negative anti-GQ1b antibodies were allowed for a diagnosis of 'probable BBE' if neurological findings were typical. Definite and probable BBE were both considered as BBE in this study. Considerable accompanying limb weakness (3 or less on the Medical Research Council scale), which has been reported as 'BBE with limb weakness'³ or 'overlapping BBE and Guillain-Barré syndrome (GBS)',¹² was defined as acceptable for BBE in our criteria.

Statistical analysis

Differences in incidence were examined using a χ^2 or Fisher's exact test. Differences in medians were examined using a Mann-Whitney U test. Differences were considered significant for $p < 0.05$ in two sided tests. Statistical calculations were made using SPSS V.19 software (IBM Japan).

RESULTS

Estimated number of cases of BBE in Japan

In the first step of the survey, a response was obtained from 1526 facilities (response rate 43.3%; 43.2% from neurology, 30.1% from internal medicine, 60.7% from paediatric and 42.2% from neurosurgery departments), of which 116 facilities had 181 cases of brainstem encephalitis (in 102 men and 79 women) for the specified 3 year period (figure 1). Based on these data, the number of patients in Japan with brainstem encephalitis was estimated as 704 (95% CI 478 to 930) over 3 years. In the second step of the survey, detailed information regarding 87 cases of brainstem encephalitis was obtained (response rate 48.1%), of which 37 (42.5%) met the diagnostic criteria for BBE (definite 19 cases; probable 18 cases). Therefore, the total number of BBE cases was estimated as $704 \times 42.5\% = 299$ for the 3 years.

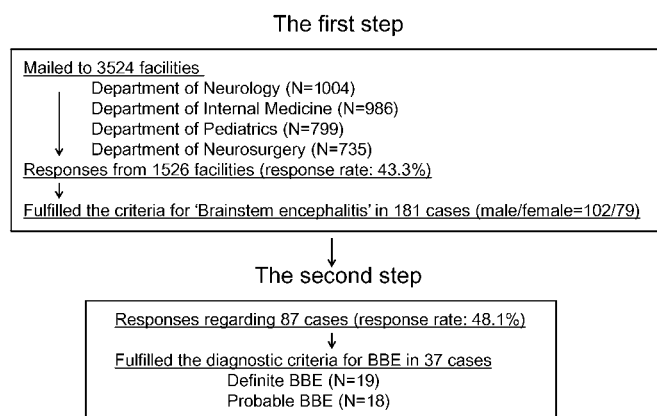


Figure 1 Flow of the survey. BBE, Bickerstaff brainstem encephalitis (diagnostic criteria for BBE are shown in box 1).

Assuming that BBE is a disease that does not require hospitalisation for more than 1 year because of good recovery, we estimate that the annual rate of BBE onset in Japan is $299/3=100$ cases.

Comparison of the diagnoses made by primary physicians and our 'in house' criteria

The second step of the survey identified 36 patients with a diagnosis of BBE made by primary physicians, among whom 31 (86%) fulfilled our own diagnostic criteria for BBE (definite and probable). In contrast, among 37 patients who met our diagnostic criteria for BBE (definite and probable), 31 (84%) patients were given a diagnosis of BBE by primary care physicians.

Characteristics of BBE cases

Cases of BBE showed a slight male predominance (male/female ratio 1.3:1) and were distributed throughout all age groups, peaking in 20–30 year olds (figure 2). The median age was significantly lower (35 years-old) than that (55 years-old) of patients with brainstem encephalitis who did not meet the diagnostic criteria for BBE ('other brainstem encephalitis (BE)') (table 1). Antecedent infectious symptoms were frequently seen (78%), especially symptoms of upper respiratory tract infections (61%). Double vision was the most frequent (27%) initial neurological symptom and the second was numbness at the distal extremities (19%); the latter was significantly more frequent in BBE than in other BE ($p=0.03$; OR=5.6; 95% CI 1.1 to 28.7). Besides the triad (external ophthalmoplegia, ataxia and impaired conscious level) of BBE criteria, characteristic neurological

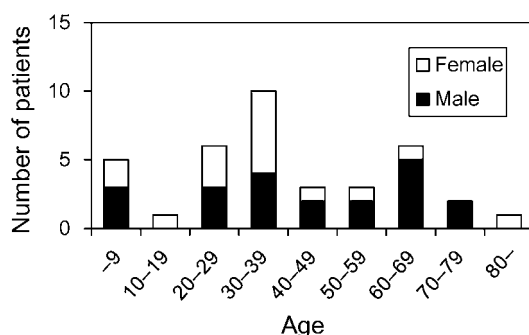


Figure 2 Age distribution of patients who met the diagnostic criteria for Bickerstaff brainstem encephalitis over 3 years (2006–2009).

features during the course of the disease were oropharyngeal palsy (62%; $p=0.03$; OR=3.0, 95% CI 1.1 to 6.2 compared to other BE), absent or decreased tendon reflex (67%; $p=0.01$; OR=3.6, 95% CI 1.3 to 9.7) and objective sensory disturbance at the distal extremities (33%; $p=0.01$; OR=6.5, 95% CI 1.6 to 26.3). It is also noteworthy that abducent predominant limitation of eye movement was more often seen in BBE patients than in those with other BE (58% vs 33%), although the difference did not reach statistical significance ($p=0.08$). Onset to the peak period was shorter in the BBE patient than in other BE patients: the patients' disabilities peaked within 1 week of onset in 23 (62%) of the 37 BBE patients. Assisted ventilation was required for 20% of these patients.

IgG anti-GQ1b antibodies were positive in 24 (75%) of the 32 patients with definite or probable BBE in whom the antibodies were examined (table 2). CSF analysis showed pleocytosis ($\geq 6/\text{mm}^3$) in 44% and increased protein ($\geq 45 \text{ mg/dl}$) in 38% of BBE patients: the latter was rarer than in other BE (67%; $p=0.02$; OR=0.31, 95% CI 0.11 to 0.85). In contrast with GBS patients, only 14% of BBE patients showed albuminocytological dissociation, although the timing of CSF sampling were not noted and therefore this rarity would merely reflect early sampling of the CSF.⁹ Detailed information regarding electroencephalography was available for 20 BBE patients, nerve conduction data for 29 and brain MRI for 35. Ten (50%) patients had abnormal electroencephalography results, mainly because of a slow wave in background activity ($n=8$). Eleven (38%) patients showed abnormal conduction in peripheral nerves, including decreased compound muscle action potential amplitude ($n=4$) or reduced F wave frequencies ($n=4$), or both. Abnormal brain MRI findings were revealed in eight (23%) patients, being significantly rarer than in other BE patients (58%; $p=0.02$; OR=0.31, 95% CI 0.11 to 0.85). Abnormal MRI findings in BBE patients were high intensity abnormalities on T2 weighted images of the brainstem, thalamus or cerebellum.

Comparison between 'definite' versus 'probable' BBE

Compared with definite BBE, probable BBE was characterised by a longer duration from onset to peak ($p=0.02$), a higher frequency of increased CSF protein (69% vs 13%; $p=0.003$) and brain MRI abnormality (39% vs 5.9%; $p=0.04$), and a rarer incidence of serology positive for anti-GQ1b antibodies (38% vs 100%; $p<0.001$) (table 1). Furthermore, marked CSF pleocytosis (cell count $51/\text{mm}^3$ or more) was revealed in five (38%) of the 13 patients with probable BBE whereas it was very rare (6.3%) in those with definite BBE. It is also noteworthy that the neurological features we found in this study to be characteristic for BBE, apart from the diagnostic triad, were more closely associated with definite BBE than probable BBE (absent or decreased tendon reflex (86% vs 50%); distal sensory disturbance (47% vs 20%); abducent predominant external ophthalmoplegia (73% vs 47%)) although the differences did not reach statistical significance ($p=0.058$, $p=0.12$ and $p=0.18$, respectively).

Choice of therapies and outcome

All BBE patients received immunological treatments, among which intravenous immunoglobulin (IVIg) ($n=34$; 92%), administration of steroid ($n=18$; 49%) or acyclovir ($n=9$; 24%) were often chosen, alone or in combination with each other, as follows: IVIg monotherapy ($n=15$; 41%), IVIg plus intravenous steroid ($n=10$; 27%), IVIg plus intravenous steroid plus acyclovir ($n=4$; 11%) and IVIg plus acyclovir ($n=3$; 8.1%).

Outcome data were available for 17 and 23 BBE patients 3 months and 1 year after onset, respectively. As shown in the

Table 1 Clinical features of patients with Bickerstaff brainstem encephalitis

	BBE				p Value† (total BBE vs other)
	Total (n=37)	Definite (n=19)	Probable (n=18)	Other* (n=50)	
Sex (M/F)	21/16	10/9	11/7	32/18	NS
Age (years) (median (range))	35 (1–87)	39 (18–78)	34 (1–87)	55 (2–84)	0.03
Antecedent infectious symptoms (n (%))	28/36 (78)	16 (84)	12/17 (71)	24/46 (52)	0.02
Fever	16/33 (49)	6/16 (38)	10/17 (59)	15/40 (38)	NS
URTI	20/33 (61)	12/17 (71)	8/16 (50)	14/44 (32)	0.01
Diarrhoea	9/33 (27)	5/16 (31)	4/17 (24)	4/41 (9.8)	NS
Initial neurological symptoms (n (%))					
Double vision	10 (27)	7 (37)	3 (17)	6 (12)	NS
Numbness at distal extremities	7 (19)	5 (26)	2 (11)	2 (4)	0.03
Conscious disturbance	7 (16)	3 (16)	4 (22)	10 (20)	NS
Dizziness	4 (11)	2 (11)	2 (11)	3 (6.0)	NS
Gait instability	3 (8.1)	3 (16)	0	6 (12)	NS
Convulsions	2 (5.4)	0	2 (11)	2 (4.0)	NS
Dysphagia/dysarthria	2 (5.4)	0	2 (11)	4 (4.0)	NS
Neurological features during the illness (n (%))					
Consciousness disturbance	34 (92)	19 (100)	15 (83)	24/46 (52)	<0.001
JCS 1–3‡	7/33 (21)	4/18 (22)	3/15 (20)	9/46 (20)	
JCS 10–30§	15/33 (46)	9/18 (50)	6/15 (40)	6/46 (13)	
JCS 100–300¶	8/33 (24)	5/18 (28)	3/15 (20)	9/46 (20)	
Optic nerve involvement	2/35 (5.7)	2/18 (11)	0/17	3/49 (6.1)	NS
External ophthalmoparesis	37 (100)	19 (100)	18 (100)	20/49 (41)	<0.001
Abducent predominance	15/26 (58)	8/11 (73)	7/15 (47)	8/24 (33)	NS
Internal ophthalmoparesis	10 (27)	7 (37)	3 (17)	9 (18)	NS
Facial palsy	15 (41)	7 (37)	8 (44)	14 (28)	NS
Oropharyngeal palsy	23 (62)	13 (68)	10 (56)	19/49 (39)	0.03
Limb weakness	19/35 (54)	12/18 (67)	7/17 (41)	18/49 (37)	NS
Tendon reflex (n (%))					
Absent/decreased	20/30 (67)	12/14 (86)	8/16 (50)	14/49 (29)	0.01
Normal	3/30 (10)	1/14 (7.1)	2/16 (13)	12/39 (31)	NS
Brisk	4/30 (13)	1/14 (7.1)	3/14 (21)	11/39 (28)	NS
Babinski/Chaddock (n (%))	12/28 (43)	5/12 (42)	7/16 (44)	12/42 (29)	NS
Ataxia	33/36 (92)	19 (100)	14/17 (82)	20/28 (71)	<0.001
Sensory disturbance (n (%))	20/36 (56)	13 (68)	7/17 (41)	16/49 (33)	NS
At distal extremities	10/30 (33)	7/15 (47)	3/15 (20)	3/42 (7.1)	0.01
Autonomic disturbance (n (%))	11/36 (31)	6/18 (33)	5 (28)	14/39 (36)	NS
Days at nadir (n (%))					
<1 week	23 (62)	15 (79)	8 (44)	22/48 (46)	0.03
1–2 weeks	10 (27)	4 (21)	6 (33)	11/48 (23)	
2–3 weeks	2 (5.4)	0	2 (11)	1/48 (2.1)	
3–4 weeks	2 (5.4)	0	2 (11)	4/48 (8.3)	
>4 weeks	0	0	0	10/48 (21)	
Hughes disability grade at nadir (n (%))					
1	1/35 (2.9)	0	1/16 (6.3)	7/44 (16)	NS
2	0/35	0	0/16	4/44 (9.1)	
3	7/35 (20)	6 (32)	1/16 (6.3)	4/44 (9.1)	
4	20/35 (57)	10 (53)	10/16 (63)	18/44 (41)	
5	7/35 (20)	3 (16)	4/16 (25)	11/44 (25)	

*Patients who met the criteria for brainstem encephalitis described in the text, but not the criteria for BBE (see box 1).

†Differences in frequencies between the groups were analysed using the χ^2 or Fisher's exact test. Differences in medians were examined by the Mann–Whitney U test.

‡The patient is awake without any stimuli, and is almost fully conscious (JCS 1), unable to recognise time place or person (JCS 2) or unable to recall name or date of birth (JCS 3).

§The patient can be aroused (then reverts to previous state after cessation of stimulation).

¶The patient cannot be aroused with any forceful mechanical stimuli.

BBE, Bickerstaff brainstem encephalitis; JCS, Japan Coma Scale; URTI, upper respiratory tract infectious symptoms.

supplementary data (available online only), 88% of BBE patients were able to walk independently within 3 months of disease onset. This good outcome was predominantly seen in patients with definite BBE, all of whom were able to walk independently 3 months after disease onset, being more frequent than that in patients with probable BBE (3/5, 60%; $p=0.074$) or other BE (18/26, 69%; $p=0.057$). One year after onset, 21 (91%) of 23 patients with BBE were able to walk independently. No associ-

ation was shown between the choice of treatment and outcome (data not shown).

DISCUSSION

This Japanese nationwide survey estimated an annual onset of 100 cases of BBE, although this number was roughly calculated based on the assumption that no patients with BBE required hospitalisation for more than 1 year after onset. Given

Table 2 Laboratory, electrophysiological and radiological findings of patients with Bickerstaff brainstem encephalitis

	BBE				p Value† (total BBE vs other)
	Total (n = 37)	Definite (n = 19)	Probable (n = 18)	Other* (n = 50)	
Serum antiganglioside antibody (IgG)					
Anti-GQ1b (n (%))	24/32 (75)	19 (100)	5/13 (38)	0/19	<0.001
Anti-GM1 (n (%))	2/33 (6.1)	1 (5.3)	1/14 (7.1)	0/18	NS
CSF					
Cell count (/mm) (median (range))	5 (1–214)	5 (1–214)	9 (2–169)	12 (0–9420)	NS
≤5/mm (n (%))	16/29 (56)	10/16 (42)	6/13 (46)	15/36 (42)	NS
6–10/mm ³ (n (%))	1/29 (3.4)	0	1/13 (7.7)	2/36 (5.6)	
11–50/mm ³ (n (%))	6/29 (21)	5/16 (31)	1/13 (7.7)	11/36 (31)	
51–100/mm ³ (n (%))	3/29 (10)	0/16	3/13 (23)	2/36 (5.6)	
>100/mm ³ (n (%))	3/29 (10)	1/16 (6.3)	2/13 (15)	6/36 (16.7)	
Protein (mg/dl) (median (range))	37 (13–400)	28 (17–72)	59 (13–400)	54 (14–549)	NS
Elevated (≥45 mg/dl) (n (%))	11/29 (38)	2/16 (13)	9/13 (69)	24/36 (67)	0.02
Albuminocytological dissociation (n (%))	4/29 (14)	1/16 (6.3)	3/13 (23)	7/36 (19)	NS
EEG abnormality (n (%))	10/20 (50)	4/8 (50)	6/12 (50)	13/23 (57)	NS
NCS abnormality (n (%))	11/29 (38)	5/15 (33)	6/14 (43)	10/20 (50)	NS
Brain MRI abnormality (n (%))	8/35 (23)	1/17 (5.9)	7/18 (39)	29/50 (58)	0.02

*Patients who met the criteria for brainstem encephalitis described in the text, but not the criteria for BBE (box 1).

†Differences in frequencies between the groups were analysed using the χ^2 or Fisher's exact test. Differences in medians were examined by the Mann–Whitney U test. BBE, Bickerstaff brainstem encephalitis; NCS, nerve conduction study.

a Japanese population of 127.7 million in 2008, we estimate that the annual incidence of BBE is approximately 0.078 per 100 000 inhabitants. Occurrence of BBE showed a slight male predominance (male/female ratio 1.3) with an average age of onset of 39 years (median 35 years), all of which are similar to figures found for a previous nationwide survey of GBS (male/female ratio 1.5; average age 39 years).¹⁷ This is the first study to provide epidemiological data for actual BBE in Japan.

This nationwide survey had some limitations: first, a relatively low responsive rate (43.3% at the first step of the survey and 48.1% at the second) and the possibility that doctors who had seen patients with brainstem encephalitis tended to actively respond to the survey. Both are unavoidable problems in questionnaire based nationwide surveys which could cause bias the estimation of the number of patients. Second, information regarding the name of the research group (BBE research group) and the member of the group, some of whom were neurologists who are actively investigating antiganglioside antibodies, was not hidden in the questionnaire. This could overestimate the number of BBE patients. On the other hand, the response rate of neurology departments was similar to that of neurosurgery departments, and lower than that of paediatric departments, although BBE is familiar among neurologists but not neurosurgeons or paediatricians. Finally, there were a considerable number of patients with brainstem encephalitis in whom serum anti-GQ1b antibodies were not tested. Our diagnostic criteria required positive anti-GQ1b antibodies for a diagnosis of probable BBE, and so patients with an incomplete neurological picture may have been missed as well as some patients with atypical BBE.

Common features at neurological, laboratory and epidemiological investigations indicate that BBE is a clinical variant of GBS.¹⁸ Therefore, comparison of the incidence of BBE with that of GBS would be interesting. Several epidemiological studies reported that the frequency of clinical variants of GBS among the total GBS population is 10.5–15.7%, including or excluding FS cases,^{19–21} but these studies have paid scant attention to BBE. We are now for the first time able to estimate the frequency of BBE among the total GBS population. A previous nationwide survey showed that the annual incidence of GBS in Japan between 1993 and 1998 was 1.15 per 100 000 inhabitants,¹⁷ and

if this frequency is unchanged, BBE is estimated to account for 6.8% of GBS. This incidence is similar to the 6% incidence of FS seen in western studies²² whereas it was reported as 25% in a Japanese study,²³ and as 18% and 19% in Taiwanese studies.^{24 25} These findings suggest that certain clinical variants of GBS predominate in Asia compared with western countries. Whether the incidence of BBE varies between countries could be determined in further studies.

A case series of 62 patients with BBE, who had been referred to a single Japanese institute for antiganglioside antibody testing, was reported to reveal clinical and laboratory features of BBE.³ However, from that study, it was not possible to identify the epidemiological data for BBE or the nosological position of BBE among all cases of brainstem encephalitis. The current study revealed that BBE accounts for 43% of brainstem encephalitis, and that among cases of brainstem encephalitis, BBE is clinically characterised by: (1) younger onset, (2) antecedent infectious symptoms, (3) onset of symptoms of numbness and objective sensory disturbance at the distal extremities during the clinical course, (4) oropharyngeal palsy, (5) abducent predominant limitation of eye movement, (6) absent or decreased tendon reflexes and (7) a shorter time from the onset to peak with good outcome, in addition to the diagnostic triad included in our diagnostic criteria. Sensory deficits at the distal extremities, with absent or decreased tendon reflexes, indicate the involvement of the peripheral nervous system, and a nerve conduction study revealed conduction abnormalities in 38% of BBE patients. Similarities with FS, including abducent predominant limitation of eye movement, good outcome and, most remarkably, frequent presence of anti-GQ1b antibodies, indicate that BBE is most likely a brainstem variant of 'anti-GQ1b IgG antibody syndrome'^{12 13} characterised not only by symptomatological features but also by the aetiological background of anti-GQ1b autoantibodies. We therefore decided to include serological evidence of the presence of IgG anti-GQ1b antibodies into the diagnostic criteria for definite BBE that were used in our study.

Previous diagnostic criteria for BBE reported by Odaka *et al*³ and Ito *et al*⁹ consisted only of neurological features with spontaneous recovery within 4 weeks of onset, in addition to

exclusive diagnosis. Because of growing evidence that BBE is mainly an anti-GQ1b antibody mediated disorder, as mentioned above, we now propose that BBE should be defined as a distinct disease entity with a specific aetiology of antiganglioside antibodies, which are not simply a symptomatological entity. Therefore, our criteria claim positive serological evidence of IgG anti-GQ1b antibodies for definite BBE and for probable BBE if neurological findings for BBE are atypical or incomplete. We showed that our criteria were valid because these sufficiently accorded with the diagnosis made by primary physicians in individual cases, and the converse was also true. A further advantage of our criteria in including the results of anti-GQ1b serology is that atypical or incomplete neurological features are acceptable for a diagnosis of probable BBE if the serology is positive. This allowed us to comparatively analyse the features of atypical BBE cases, which were classified as probable BBE using our criteria, as described below.

We found that the cases of probable BBE, in which neurological features were atypical or anti-GQ1b antibodies were not present, often had abnormally elevated protein levels and marked pleocytosis in the CSF, and abnormal MRI findings in the brain in addition to a longer duration from onset to the peak of neurological symptoms, sometimes with considerable residual deficits. We did not ask the timing of CSF sampling and therefore a low frequency of abnormally elevated protein levels in CSF would merely reflect early sampling of the CSF. However, it was reported that the frequency of CSF pleocytosis in BBE did not differ between the first (32%) and second weeks (31%),⁹ suggesting that data unavailability regarding the timing of CSF sampling does not influence the higher frequency of CSF pleocytosis in probable BBE group. Although a negative finding for the presence of anti-GQ1b antibodies might suggest the presence of undetermined antiganglioside antibodies in some cases of BBE, such as those of FS without anti-GQ1b antibodies,²⁶ these findings might also suggest that other pathogeneses, such as direct infection by an unidentified virus or an underdiagnosed inflammatory disease status, appear to play a role in the development of the 'BBE' phenotype in a considerable number of cases of probable BBE. We believe that BBE is a disease entity that consists of typical cases of BBE (definite BBE in this study) characterised by the presence of anti-GQ1b antibodies and good recovery in the majority, and of atypical cases of BBE (a component of the probable BBE in this study) characterised by a delayed recovery resulting from other pathogeneses in the minority.

In conclusion, we have provided epidemiological data for the incidence of BBE in Japan for the first time. BBE is a rare disorder, but accounts for a major proportion of brainstem encephalitis and is a minor clinical variant of GBS. Typical BBE is characterised by an FS-like clinical presentation of neurological and serological (anti-GQ1b antibodies) features with good recovery, whereas atypical BBE may be a distinct disorder, possibly with another pathogenesis. We now propose that BBE should be defined as a distinct disease entity of the brainstem variant of GBS/FS.

Acknowledgements We thank all of the physicians who participated in the survey. We also thank Professor Nobuhiro Yuki (Departments of Microbiology and Medicine, National University of Singapore) for critical reading of the manuscript, and Ms Sawako Hanaoka (Yamaguchi University Graduate School of Medicine) for her technical assistance with the survey.

Contributors MK: study design, construction of the diagnostic criteria, data collection and analysis, and drafting of the manuscript; SK, KK and TKo: constructing the diagnostic criteria, data analysis and revision of the manuscript; RU and YN: data

collection and analysis, and review of the manuscript; TKa: study design, construction of the diagnostic criteria, data collection and analysis, and review of the manuscript.

Funding This work was supported by a research grant for Bickerstaff Brainstem Encephalitis (to MK, SK, YN and TKa) from the Ministry of Health, Labour and Welfare of Japan, by a research grant for neuroimmunological diseases (to SK, TKo and TKa) from the Ministry of Health, Labour and Welfare of Japan and by a Grant-in-Aid for Scientific Research (C) (KAKENHI 20590446 to MK) from the Ministry of Education, Culture, Sports, Science and Technology of Japan.

Competing interests None.

Ethics approval This study was conducted with the approval of the institutional ethics committee at Yamaguchi University Hospital.

Provenance and peer review Not commissioned; externally peer reviewed.

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